HEALTHBRIDGE RAG CHATBOT: TPM PLAN

Technical Project Manager Execution Schedule & Comprehensive Delegation Matrix

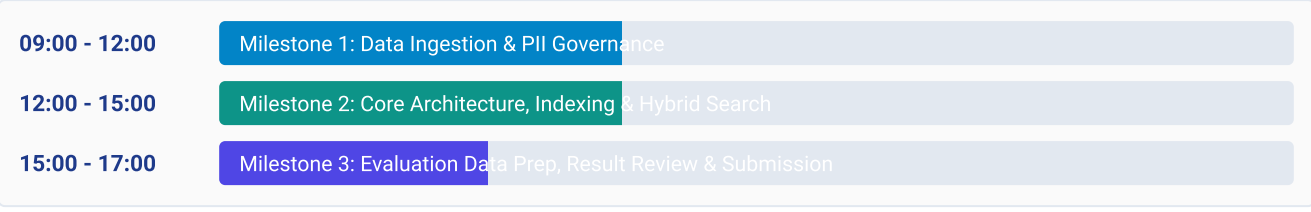
Project Manager: Shabbar AI
Role Focus: Pure PM & Governance (No Coding)
Team Size: 7 Members • 8-Hour Residency Execution

PURE GOVERNANCE FOCUS
Shabbar AI operates as full-time Technical Project Manager, coordinating velocity, dependencies, board management, and quality gates.

STRUCTURED REVIEW GATES
PM conducts official quality checkpoints and review pairings at every milestone to audit vector indices, logs, and safety rules.

REPORT & README OWNER
Direct ownership of administrative deliverables, co-authoring the `01_ingestion` report and publishing final project documentation.

1. HIGH-LEVEL 8-HOUR EXECUTION SCHEDULE



2. DETAILED DELEGATION MATRIX (MILESTONES 1 & 2)

Milestone & Time Window	Team Members Executing	Detailed Member Task Breakdown	Shabbar's TPM Management & Actions
Milestone 1 09:00 - 12:00 *Data Ingestion & PII Governance*	Zaineb Syeda Ravalika Gundu Shabbar AI (PM)	• **Zaineb Syeda:** Executes Stage 1 web crawler adhering strictly to `robots.txt` limits (≤1 rps rate limits) and sitemap seeds. • **Ravalika Gundu:** Handles Stage 2 data extraction, pulling clean HTML text via `trafilatura` and structured tables/text from PDFs via `pdfplumber`.	• Initialize GitHub project board, assign stage issues, and enforce branch protection. • PM GATE 1 Review and sign off on Ravalika's extraction PR (review pairing) and audit ingestion/PII logs. • Co-author `01_ingestion` report.
Milestone 2 12:00 - 15:00 *Core Architecture, Indexing & Hybrid Search*	Abul Hasan Harsha Vardhan Sai Chandra Manish Karki Shabbar AI (PM)	• **Abul Hasan:** Manages overall system architecture and implements Stage 3/6 structure-aware chunking (`sa-v1`). • **Harsha Vardhan:** Builds Stage 4 `bge-base-en-v1.5` dense embeddings and configures persistent ChromaDB vector store. • **Sai Chandra:** Implements Stage 7 BM25 lexical search and Reciprocal Rank Fusion (RRF) algorithm. • **Manish Karki:** Configures Stage 8 cross-encoder reranking (`bge-reranker-base`).	• Monitor cross-stage team dependencies and maintain velocity toward noon B2 baseline deadline. • PM GATE 2 Audit index health metrics (`index_card.json`) and verify baseline system functionality.

3. MILESTONE 3 EXECUTION MATRIX (15:00 - 17:00)

Milestone & Time Window	Team Members Executing	Detailed Member Task Breakdown	Shabbar's TPM Management & Actions
Milestone 3 15:00 - 17:00 *Evaluation Data Prep, Result Review & Submission*	**Zaineb Syeda** **Ravalika Gundu** (Data Sci) **Manish Karki** (Data Sci) **Abul Hasan** (Architect) **Shabbar Al (PM)**	• **Zaineb Syeda:** Implements Stage 9 generation (`gpt-4o-mini`), interactive Streamlit chat UI, and emergency / prompt-injection safety guardrails. • **Data Scientists (Ravalika & Manish):** Manually prepare, verify, and cross-check evaluation golden set (`eval/golden_set.jsonl`) via human/manual inspection of live UCSF pages for verified ground-truth answers. • **Model Accuracy Reviewers (Shabbar & Hasan):** Review chatbot output results against real verified answers to evaluate model accuracy, grounding quality, and system performance.	• Oversee final integration of Streamlit UI, safety guardrails, and evaluation harnesses. • **PM GATE 3** Finalize `01_ingestion` report and project README, verify all PR approvals, and lock repository for final submission.

4. RESPONSIBILITIES SUMMARY BY TEAM MEMBER

Shabbar Al (Technical Project Manager)
Pure management & governance, GitHub board setup, review pairings, 3 PM quality gates, co-authoring `01_ingestion` report & README, and model accuracy result reviews.

Abul Hasan (Solutions Architect)
Overall system architecture, Stage 3/6 structure-aware chunking (`sa-v1`), and model accuracy evaluation co-lead with Shabbar.

Ravalika Gundu (Data Scientist)
Stage 2 HTML/PDF data extraction via `trafilatura` and `pdfplumber`, plus manual curation/verification of the golden evaluation dataset.

Manish Karki (Data Scientist)
Stage 8 cross-encoder reranking (`bge-reranker-base`), evaluation harness execution, and golden set manual ground-truth verification.

Harsha Vardhan (Backend Engineer)
Stage 4 dense embedding generation using `bge-base-en-v1.5` and persistent ChromaDB vector store setup.

Sai Chandra (Backend Engineer)
Stage 7 lexical search using BM25 and implementation of Reciprocal Rank Fusion (RRF) for hybrid retrieval.

Zaineb Syeda (Frontend Engineer)
Stage 1 compliant web crawler execution, Stage 9 `gpt-4o-mini` generation, interactive Streamlit chat UI, and emergency/prompt-injection safety guardrails.

5. EXECUTIVE PRESENTATION TALKING POINTS FOR PROFESSOR

- **Pure Management Focus:** "I serve strictly as Technical Project Manager, steering project boards, PR approvals, branch protection, and milestone velocity without writing baseline code."
- **Rigorous Delegation:** "Every single technical task—crawler limits, PDF extraction, vector embeddings, reranking, and safety guardrails—is mapped to specific engineering owners."
- **Data Quality & Accuracy Ownership:** "Our Data Scientists manually verify ground-truth source data, while Hasan and I independently audit chatbot generated outputs against real-world answers for factual accuracy."